

Mr. Deo Raj Biswa

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Education

- Master of Engineering in Mechatronics – Asian Institute of Technology, Thailand (2020)
- Postgraduate Diploma in Higher Education – Samtse College of Education (2018)
- Bachelor of Technology – EILM University, India (2013)
- Higher Secondary (Science) – Punakha HSS, Bhutan (2008)

Academic and Professional Experience

- Head of Department, Electrical and Electronics Engineering – JNEC (July 2024 – Present)
- Programme Leader for Diploma in Electronics and Communication Engineering (July 2024–Present).
- Lecturer (Assistant & Associate Levels) – JNEC (2014 – Present)
 - Taught: Industrial Automation, Process Control, Embedded Systems, Mechatronics, Control Systems, Programming (C, Embedded-C, Assembly).
 - Developed new modules and Diploma in EV Technology.
 - Led Electric Dryer development (WIPO collaboration).
 - Supervised final year student projects.
 - Committee roles: Chair of PBoE, Industrial Attachment Coordinator, Academic/Sports Committees, etc.
 - Center Head and Coordinator for Center of Appropriate Technology and TISC – JNEC (2022 – 2024)
 - Coordinated and Organized: BIFA International Workshop, BUSF (2014, 2022)

Scientific Contribution/Membership

- **Reviewer:**
 - *Engineering Applications of Artificial Intelligence* (Elsevier) – Reviewed manuscripts in intelligent systems and embedded AI applications (2024–present)
 - *Journal of Computer Sciences and Informatics* – Reviewed manuscripts (2025)

Research Projects and Collaborations

- Electric Dryer Development – WIPO (Lead)
- Grassroot STEM Education – UNICEF-Bhutan (Co-Investigator)
- STEM Outreach – REF Project with Samtse College (Co-Investigator)
- Biometric Electoral System – AURG Funded (Co-Investigator)
- Forest Fire Mapping – CARG Funded (Co-Investigator)
- Waste Stabilization using IVC – BTFEC Funded (Co-Investigator)

Thesis/Project

- Development of Multipurpose Electric Dryer, 2023
- Development of Segway Forklift, 2020
- Development Data Logger, 2013

Awards, Grants, and Fellowships

- Bhutan-India Friendship Workshop on Climate Action: AI, IoT, Robotics, and Artistic Fusion for a Sustainable Indo-Bhutan – 2025
- Bhutan India Friendship Workshop on Applications of Artificial Intelligence in Renewable Energy, EV Charging, and Grid Modernization infused with GNH Values – 2024
- RCSC Civil Service Award – Bronze (2024)
- Erasmus+ Mobility Grant – Sofia Technical University (2024)
- ADB-JSP Scholarship – Mechatronics (2018)
- CARG, AURG, REF, UNICEF, BTFEC Grants (2016–2024)

Publications and Conference Papers

- Journal Articles:
 - 3D Concrete Printer (2025)
 - Building Self-confidence (2025)
 - Automated Wheelchair with Fall Detection (2023)
 - Energy Efficient Lighting Design (2023)
 - STEM Education in Grassroots of Bhutan (2023)
 - Biometric Electoral System (2021)
- Conference Papers:
 - Smart Car Parking – JNEC Thruel Rigsar Toed
 - Smart Metering – JNEC Thruel Rigsar Toed
- Workshops and Presentations:
 - STEM Arduino Training for Teachers – 2023
 - Development of Segway-type Forklift – 5th Faculty Seminar, 2022
 - Winter STEM Camps – 2021, 2022

Articles and Outreach

- Mini-project – An Innovative Display (2022)
- The Undisclosed Fate (2022)
- My Career Journey as an Educator (2022)

Technical Skills

- Programming: C, Embedded-C, Assembly, Python
- Tools: Arduino IDE, MATLAB, Keil, Multisim, SolidWorks.
- SCADA/PLC: TIA Portal, Winpro Loader, Intouch

- Hardware: PC troubleshooting, installation
- Counseling: Life skills and sexuality education

Personal Information

- Date of Birth: 16 July 1989
- Gender: Male
- Marital Status: Married
- CID No.: 11811002959
- Permanent Address: Zamdongkhar Bapegarkha, Guma Gewog, Punakha